

Differentiate Between Git and GitHub

Git	GitHub
Git is a distributed version control system	GitHub is a cloud-based hosting platform for Git repositories
Installed locally on a system	Web-based platform
Used to track code changes	Used to store, share, and collaborate on Git repositories
Works offline	Requires internet for collaboration features
Open-source tool	Company/platform built on top of Git

What is Version Control?

Version Control is a system that helps track changes made to files over time, allowing developers to manage, compare, and restore different versions of code. It enables multiple developers to work on the same project efficiently without conflicts.

Scope of Version Control

The scope of version control includes:

- **Tracking code changes and history**
 - **Collaboration among multiple developers**
 - **Backup and recovery of code**
 - **Branching and merging features**
 - **Maintaining different versions (releases)**
 - **Auditing and accountability (who changed what and when)**
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Use Cases of Version Control

- **Team collaboration on software projects**

- **Maintaining source code history**
 - **Bug tracking and rollback to stable versions**
 - **Managing releases and deployments**
 - **Supporting CI/CD pipelines**
 - **Open-source project development**
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Compare Git Branch vs Git Merge

Git Branch

Used to create a separate line of development

Helps work on features independently

Lightweight and fast

Improves parallel development

Git Merge

Used to combine changes from one branch into another

Integrates completed work

Can create merge commits

Ensures code integration

Summary:

Git manages source code versions locally, while GitHub enables collaboration and hosting. Version control improves collaboration, stability, and traceability in modern software development.